

EDGE MINDS HACKATHON 2026

Team Name : Electro_Neurons

Team ID : EDGM26-DWDPMR

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Problem Statement :

Build an AI agent where the SLM is the brain - planning steps, calling tools, evaluating results, and self-correcting. Minimum 3 custom tools and multi-step execution. Apply to research assistants, data analysis, DevOps monitoring, or personal productivity

Solution / Project Proposal :

Abstract & Problem

Edge devices need low-latency, offline monitoring, but current solutions are either fragile rule-based scripts or cloud-dependent agents with latency and privacy issues. We build a fully on-device agent that runs within a 1.2B-parameter budget using a lightweight SLM like Llama-3.2-1B via Ollama.

Solution Overview

Brain: Local $\approx$ 1B SLM (Llama-3.2-1B) running through Ollama, optimized for edge hardware (4–8 GB RAM).

Agent Loop: Minimal Python ReAct loop (reason $\rightarrow$ tool call $\rightarrow$ observation $\rightarrow$ next step) with simple regex parsing, no heavy frameworks.

Custom Tools (local, easy to implement):

get_system_status() – CPU, RAM, disk, thermals via shell commands.

scan_local_directory(path) – detect missing configs, large files, log bloat.

save_edge_log(payload) – append structured diagnostics to a local log file.

Self-Correction

If a tool call fails, the Python layer captures the error and feeds it back to the SLM as an Observation. The model then adjusts its plan (e.g., fixes arguments, tries fallback paths) and issues a corrected tool call. This creates an offline, self-healing edge agent that is simple to implement yet showcases real multi-step, tool-using, and self-correcting behavior within a strict 1.2B parameter budget.